



△ DRY FALLS

The Palouse River flows through a giant gorge that was carved out by the Missoula floods.

▽ ISOSTATIC REBOUND

As North America's ice sheets melted, the weighed-down land beneath slowly began to rise—a process called isostatic rebound. This effect can be seen, for example, in the exposure of old shorelines.

▽ CRACKS IN THE LAND

The San Andreas faultline visibly slices across the Mojave Desert, dividing the Pacific and North American tectonic plates.



Mountains rose as the submerged oceanic plate melted to create volcanoes. Islands slammed against the continent, and the collisions piled up layers of rock on the continent's coast. The Rocky Mountains are unusual in that they rose up farther inland than coastal mountain ranges, such as the Andes in South America, normally do. The reason remains a mystery for geologists—it may be that the ocean plate descended at a much flatter angle and so melted in the mantle at a point farther beyond the coast.

Arctic Ocean had permanent sea ice.

The ice sheet's massive weight gouged deep scars into the land that would later become basins for the Great Lakes, and a huge lake called Lake Agassiz was ponded up against it. As the giant ice sheet finally melted, it unleashed cataclysmic torrents called the Missoula floods—in Washington and Oregon—carving out giant canyons and vast waterfalls that are now left as huge, bare cliffs, such as those seen at Dry Falls, Washington.

ICE AGE

During the coldest period of the last ice age 24,500 years ago, a giant ice sheet covered the northern half of North America, and the Arctic Ocean had permanent sea ice.



American icebox

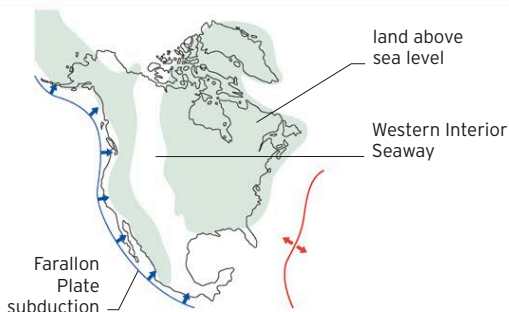
There have been five major ice ages in Earth's history. The most recent began 1.8 million years ago and lasted until just 11,700 years ago. During that time, much of Canada and the northern U.S. were covered by a vast ice sheet, and most of the

Active faultline

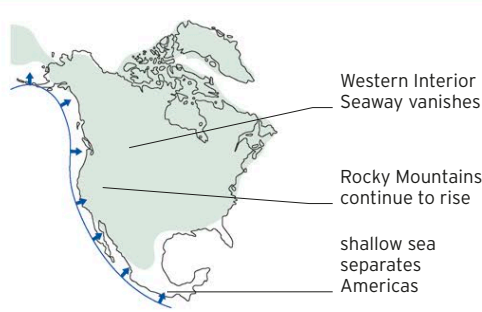
The small Farallon Plate was once sandwiched between the Pacific and North American Plates. It was entirely swallowed up 30 million years ago as the two bigger plates collided and subducted. When the subduction stopped, the two gigantic plates began sliding past each other rather than smashing together. It is here that the world's most famous strike-slip fault formed—the San Andreas Fault. The fault is 810 miles (1,300 km) long and stretches from the Mendocino coast south to the San Bernardino Mountains and the Salton Sea. Since it formed, the land on either side has moved at least 340 miles (550 km). Every juddering movement unleashes devastating earthquakes.

BUILDING THE NORTH AMERICAN CONTINENT

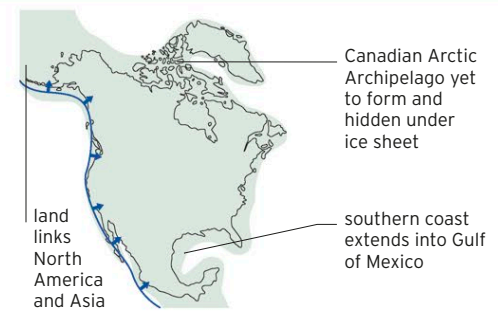
KEY — Convergent boundary — Divergent boundary



94 MILLION YEARS AGO The Western Interior Seaway, a large inland sea, splits North America in half. It was created as the subduction of the Farallon Plate pulled the interior of the continent down.



50-40 MILLION YEARS AGO As the Rocky Mountains rise higher and higher, the Western Interior Seaway is reduced to shallows and eventually dries up.



18,000 YEARS AGO With huge quantities of water locked up in the ice sheets, sea levels are at times so low that much of North America's coastline extends beyond today's boundaries.